

Explainable Real Estate Analytics: A Comparative Analysis of Machine Learning Frameworks for House Price Prediction

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Abstract

Abstract—The real estate industry generates massive volumes of heterogeneous, spatial, and structural data critical for accurate macroeconomic valuation, municipal planning, and individual investment decision-making. Conventional property appraisal techniques depend heavily on localized human heuristics, comparative market analysis (CMA), and traditional hedonic linear regressions, which frequently fail to capture complex non-linear feature interactions, spatial heteroscedasticity, and volatile macroeconomic shifts. This research presents a rigorous, end-to-end explainable machine learning framework for real estate price forecasting that systematically integrates robust data cleaning, multi-dimensional exploratory data analysis (EDA), advanced feature engineering, multi-model benchmarking, and post-hoc interpretability. We evaluate four primary model paradigms: Decision Tree Regressor, Random Forest Regressor, eXtreme Gradient Boosting (XGBoost), and Categorical Boosting (CatBoost), benchmarked against traditional baseline models (Ordinary Least Squares and Support Vector Regression) across three multi-regional benchmark datasets. Model performance is comprehensively assessed via Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Coefficient of Determination (R^2), and Mean Absolute Percentage Error (MAPE), paired with non-parametric Wilcoxon signed-rank hypothesis testing. CatBoost achieved superior predictive accuracy with an R^2 of 0.9142 and a lowest RMSE of \$24,185. To demystify black-box predictions, we implement dual-level explainability through SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), providing both global macroeconomic feature rankings and faithful local instance-level transaction breakdowns. The framework bridges the critical gap between raw algorithmic predictive power and regulatory, trust-driven stakeholder transparency in modern real estate analytics.

Keywords—Real Estate Analytics, House Price Prediction, Machine Learning, Gradient Boosting, CatBoost, XGBoost, Model Explainability, SHAP, LIME, Feature Engineering.

1. Introduction

The real estate market represents one of the primary pillars of global economic stability and individual wealth accumulation. Accurate property valuation is fundamental not only to individual buyers and sellers, but also to commercial banking institutions, mortgage underwriters, municipal taxation authorities, and urban development planners [1, 2]. However, real estate valuation is inherently complex due to the multidimensional nature of residential properties, which encompass physical architectural attributes (e.g., square footage, bedroom/bathroom counts, construction quality), structural age, temporal market cycles, geographic micro-locations, and socio-economic environmental factors [3, 4].

1.1 Background

Historically, real estate valuation has relied heavily on traditional comparative market analysis (CMA) conducted by human appraisers alongside standard econometric Hedonic Price Models (HPM) rooted in Ordinary Least Squares (OLS) regression [3, 5]. While these foundational methods provided transparent coefficients, they inherently assume linear, parametric relationships, homoscedastic residual distributions, and feature independence [6]. In reality, real estate markets exhibit severe spatial autocorrelation, multicollinearity, and non-linear interactions; for example, the marginal value of an additional bathroom varies significantly depending on total living area, neighborhood socioeconomic tier, and property age [7].

The advent of data science and statistical learning paradigms has enabled the processing of expansive, multi-attribute transaction datasets. Non-parametric machine learning algorithms—ranging from ensemble bagging architectures like Random Forests to gradient boosting frameworks like XGBoost, LightGBM, and CatBoost—have consistently demonstrated superior predictive power over linear baselines by capturing high-degree feature interactions without rigid distributional assumptions [8–10].

1.2 Problem Statement

Despite substantial improvements in raw empirical predictive metrics, the adoption of machine learning in real-world real estate appraisal remains severely constrained by two fundamental bottlenecks:

- 1. The Predictive-Interpretability Trade-off:** High-capacity non-linear ensembles operate as opaque “black boxes”. In high-stakes financial domains involving multi-decade mortgage underwriting, regulatory compliance, and anti-discriminatory lending legislation (e.g., the Equal Credit Opportunity Act), opaque predictions cannot be readily audited or trusted by institutional stakeholders [11, 12].
- 2. Suboptimal Feature Representation and Inconsistent Preprocessing Pipelines:** Property datasets often suffer from high cardinal categorical variables (e.g., zip codes, neighborhood clusters),

non-Gaussian skewed target distributions, unhandled outliers, and missing structural features. Without structured exploratory data analysis (EDA), domain-guided feature transformations, and rigorous statistical validation, models frequently overfit to local sampling artifacts [13, 14].

1.3 Research Gap

While numerous studies have applied individual regression models to house price datasets, the existing literature exhibits critical methodological gaps:

- Most prior works evaluate models on isolated, single-city datasets without multi-dataset validation across differing regional market dynamics.
- Machine learning benchmarks frequently omit systematic ablation studies that decouple the individual contributions of non-linear target transformations, spatial feature engineering, and categorical encoding mechanisms.
- Post-hoc Explainable Artificial Intelligence (XAI) is frequently treated as an afterthought, often presenting basic Gini feature importance without unified game-theoretic global (SHAP) and perturbation-based local (LIME) validation across contrasting individual property profiles.

1.4 Research Objectives

To address these limitations, this research presents a holistic, explainable machine learning framework for residential real estate price valuation. The primary objectives are:

1. To design and execute a comprehensive end-to-end data preparation and Exploratory Data Analysis (EDA) pipeline that systematically handles missing values, detects and mitigates multivariate outliers, addresses multicollinearity, and performs domain-driven feature engineering.
2. To implement, hyperparameter-tune, and comparatively benchmark four tree-based machine learning paradigms (Decision Tree Regressor, Random Forest Regressor, XGBoost, and CatBoost) against standard linear (OLS) and kernelized (SVR) baseline models across multi-regional datasets.
3. To rigorously evaluate model performance across multiple complementary metrics (MAE, MSE, RMSE, R^2 , and MAPE) alongside rigorous non-parametric statistical hypothesis testing.
4. To integrate dual-level model interpretability using unified Kernel/Tree SHAP and LIME, thereby validating both global economic determinants and granular, instance-level valuation explanations.

1.5 Research Questions

This investigation is guided by four central research questions:

- **RQ1:** How do modern gradient boosted decision trees (XGBoost, CatBoost) compare against bagging architectures (Random Forest) and baseline regressors in predictive precision and computational

training efficiency across heterogeneous real estate datasets?

- **RQ2:** To what extent does domain-driven feature engineering (e.g., spatial clustering, age decay interaction, total living-to-lot ratios) enhance the empirical R^2 and reduce residual variance?
- **RQ3:** How do SHAP and LIME explanations reconcile with classical hedonic price theory, and do they reliably expose non-linear threshold effects in property valuation?
- **RQ4:** Does CatBoost's native ordered target statistic encoding provide a statistically significant advantage over standard one-hot/ordinal encoding in high-cardinality geographic features?

1.6 Contributions

The primary contributions of this work are summarized as follows:

- **Comprehensive Comparative Benchmark:** A multi-metric, multi-dataset empirical evaluation of six distinct regression paradigms under standardized 5-fold cross-validation.
- **Domain-Specific Feature Engineering Pipeline:** Formulation of derived structural, temporal, and spatial composite indicators that systematically boost model explanation fidelity and variance explained.
- **Dual-Perspective Explainable AI Architecture:** A unified framework combining global SHAP summary distributions, interaction dependence plots, and localized LIME surrogate breakdowns, transforming opaque ensembles into auditable valuation tools.
- **Statistical Rigor and Ablation Analysis:** Inclusion of Wilcoxon signed-rank hypothesis tests and component-wise ablation studies demonstrating the exact performance gains attributable to each preprocessing and algorithmic stage.

1.7 Organization of the Paper

The remainder of this manuscript is organized as follows: Section 2 reviews foundational and contemporary literature across classical econometrics, machine learning, deep learning, and explainability. Section 3 delineates the proposed architectural framework, preprocessing protocols, feature engineering formulations, and mathematical foundations of the models and XAI methods. Section 4 outlines the hardware/software setup, validation split strategies, metric formulations, and cross-validation procedures. Section 5 presents comprehensive empirical results, ablation studies, and explainability visualizations. Section 6 interprets key findings, contrasts them with prior literature, and discusses practical/theoretical implications. Section 7 addresses methodological constraints, and Section 8 concludes the paper with future research directions.

2. Related Work

Property valuation has evolved across four distinct methodological eras: hedonic econometric modeling, spatial econometrics, non-parametric machine learning, and explainable artificial intelligence.

2.1 Existing Approaches: Hedonic Regressions

The economic foundation of real estate pricing was formalized by Rosen’s Hedonic Pricing Model (HPM) [3], which posits that goods are valued for their utility-bearing characteristics. Early real estate research applied Ordinary Least Squares (OLS) regression to estimate marginal implicit prices for individual structural attributes [2, 5]. However, OLS models suffer from severe limitations: they fail to handle spatial autocorrelation, assume linear or log-linear functional forms, and are highly sensitive to multicollinearity among correlated architectural dimensions [4, 6]. Subsequent spatial econometric models, such as Geographically Weighted Regression (GWR) [15] and Spatial Autoregressive Models (SAR) [16], improved spatial adaptation but remained computationally intensive and struggled with high-dimensional feature spaces.

2.2 Machine Learning Approaches

The transition toward non-parametric machine learning addressed many of the rigid functional assumptions of hedonic regressions. Early studies demonstrated that Artificial Neural Networks (ANN) and Support Vector Machines (SVM) could outperform OLS regressions in predicting property transaction values [17, 18].

Subsequent breakthroughs centered on ensemble learning. Breiman’s Random Forests (RF) [8] introduced bagged decision trees that mitigated variance and effectively captured complex interactions [7]. With the advent of Gradient Boosting Machines (GBM) [19] and Chen and Guestrin’s XGBoost [9], boosting algorithms emerged as the state of the art for tabular data. Recent benchmarks across Kaggle housing competitions and academic property datasets confirm that tree-based gradient boosting architectures consistently outperform standard regression models in terms of RMSE and R^2 [10, 14, 20]. CatBoost, introduced by Prokhorenkova et al. [10], further advanced tabular modeling by handling categorical features without preprocessing-induced target leakage via ordered boosting.

2.3 Deep Learning Approaches

Recent investigations have explored deep neural network architectures for real estate analytics. Multimodal deep learning frameworks integrate structured tabular property characteristics with unstructured satellite imagery, street-view photographs, and textual property descriptions [21, 22]. While deep neural architectures excel at multimodal integration, their performance on purely tabular housing datasets often remains on par with or marginally inferior to well-tuned gradient boosted decision trees (GBDTs), while incurring substantial computational overhead and severe interpretability challenges [23].

2.4 Explainability and Interpretability Techniques

The opacity of high-performing machine learning models has catalyzed the development of Post-hoc Explainable

AI (XAI) frameworks. Ribeiro et al. [24] introduced Local Interpretable Model-agnostic Explanations (LIME), which builds sparse local linear surrogates around specific individual predictions to provide intuitive feature perturbations. Lundberg and Lee [25] proposed SHapley Additive exPlanations (SHAP), rooted in cooperative game theory [26], which calculates unique, additive feature attribution values satisfying desirable mathematical properties: efficiency, symmetry, dummy, and additivity. Recent works in financial econometrics and urban analytics have begun adopting SHAP to extract reliable feature importance and identify localized non-linear thresholds [14, 27].

2.5 Critical Comparison

Table 1 provides a systematic critical comparison of representative literature across methodological paradigm, interpretability depth, spatial modeling capability, and predictive performance.

2.6 Identified Research Gap in Existing Literature

Despite significant advances, current research frequently lacks an integrated workflow that couples advanced gradient boosting frameworks with rigorous statistical significance testing, domain-informed engineering of derived physical and socio-spatial indicators, and dual-perspective (global-macro and local-micro) explainability. This research directly bridges this gap.

3. Proposed Methodology

3.1 Overall Framework

The proposed real estate analytical pipeline comprises five sequential modules: (1) Data Ingestion and Quality Assessment, (2) Multi-angle Exploratory Data Analysis, (3) Feature Engineering and Dimensionality Optimization, (4) Model Training and Hyperparameter Optimization, and (5) Dual-Perspective Explainability and Interactive Visualization.



Figure 1: System architecture of the proposed analytical framework.

3.2 Datasets

To ensure robust generalization, the methodology was evaluated across three distinct benchmark housing datasets:

1. **Dataset 1 (Kaggle House Price Benchmark [28]):** Contains transactional records with primary structural attributes including living area, lot size, bedroom/bathroom distributions, building condition, and sale price.

Table 1: Critical Comparison of Literature in Real Estate Valuation and Machine Learning

Study	Core Methodology	Interpretability Approach	Metrics	Critical Findings & Limitations
Rosen (1974) [3]	Hedonic Linear Regression (OLS)	Inherent Parametric Co-efficients	R^2 , F -stat	Foundational theory; fails on non-linear spatial interactions.
Bourassa et al. (2010) [7]	OLS vs. Spatial GAM vs. Neural Net	Statistical significance (p -values)	Out-of-sample RMSE	Identified spatial non-linearities; high computational complexity.
Park & Bae (2015) [20]	C4.5 DT, RIPPER, Naive Bayes, AdaBoost	Global Decision Tree Rules	Accuracy, RMSE	Demonstrated rule-based modeling; lacked modern gradient boosting.
Kok et al. (2017) [14]	Random Forest & Spatial Smoothing	Gini Impurity Feature Importance	R^2 , MAPE, Median	Large city validation; Gini importance biased toward continuous features.
Law et al. (2019) [21]	Multimodal Deep CNN + Tabular	Grad-CAM Heatmaps	RMSE, R^2	Effectively used street imagery; opaque tabular interactions.
Our Framework	DT, RF, XGBoost, Cat-Boost + SVR/OLS	Dual Global/Local + LIME	XAI: SHAP, MAE, RMSE, R^2, MAPE + Wilcoxon	Unified explainability, rigorous feature engineering, and statistically validated multi-model benchmark.

2. **Dataset 2 (Multi-Feature Urban Housing Dataset [29]):** Encompasses high-density metropolitan transaction records with detailed architectural quality ratings, garage capacities, and basement finishing metrics.

3. **Dataset 3 (House Prices India Regional Dataset [30]):** Captures dynamic regional sub-market properties across Indian urban agglomerations, incorporating localized amenities, proximity metrics, and dynamic geographic coordinates.

3.3 Data Preprocessing

Let the raw housing dataset be defined as $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^D$ represents the D -dimensional feature vector and $y_i \in \mathbb{R}^+$ represents the continuous transaction sale price.

3.3.1 Missing Value Imputation

Missing values in categorical attributes $\mathbf{x}^{(cat)}$ are imputed using mode substitution or assigned an explicit missing indicator category “None”. Continuous structural features $\mathbf{x}^{(cont)}$ are imputed using median substitution conditioned on neighborhood clusters $\mathcal{C}_k$:

$$\hat{x}_{i,j}^{(cont)} = \text{median} \left(\{x_{m,j}^{(cont)} \mid m \in \mathcal{C}_k\} \right) \quad \text{for } i \in \mathcal{C}_k \quad (1)$$

3.3.2 Outlier Detection and Treatment

Extreme outliers in floor areas and transaction prices were identified using the Interquartile Range (IQR) rule:

$$\text{IQR}_j = Q_3(x_j) - Q_1(x_j) \quad (2)$$

Values lying outside $[Q_1(x_j) - 1.5 \cdot \text{IQR}_j, Q_3(x_j) + 1.5 \cdot \text{IQR}_j]$ were winsorized to the 1st and 99th percentiles to preserve sample size while suppressing gradient instability during boosting optimization.

3.3.3 Target Variable Transformation

Because real estate prices exhibit right-skewness and heteroscedasticity, a logarithmic transformation is applied:

$$z_i = \ln(y_i) \quad (3)$$

Transforming the target variable stabilizes residual variance across varying price scales and normalizes error distributions.

3.4 Feature Engineering

To capture complex physical and architectural interactions, we formulate several domain-specific derived features:

1. Total Usable Area (A_{total}):

$$A_{\text{total}} = A_{\text{living}} + A_{\text{basement}} + 0.5 \cdot A_{\text{garage}} \quad (4)$$

2. Effective Structural Age (τ_{eff}):

$$\tau_{\text{eff}} = T_{\text{sold}} - \max(T_{\text{built}}, T_{\text{remodel}}) \quad (5)$$

3. Bathroom-to-Bedroom Ratio ($R_{\text{bath/bed}}$):

$$R_{\text{bath/bed}} = \frac{N_{\text{full.bath}} + 0.5 \cdot N_{\text{half.bath}}}{N_{\text{bedroom}} + \epsilon} \quad (6)$$

4. Spatial Density Index (S_{density}):

$$S_{\text{density}} = \frac{A_{\text{living}}}{A_{\text{lot}} + \epsilon} \quad (7)$$

5. Composite Quality Score (Q_{comp}):

$$Q_{\text{comp}} = w_1 \cdot \text{OverallQual} + w_2 \cdot \text{OverallCond} \quad (8)$$

3.5 Feature Selection

Multicollinearity was diagnosed via the Variance Inflation Factor (VIF):

$$\text{VIF}_j = \frac{1}{1 - R_j^2} \quad (9)$$

Features exhibiting $\text{VIF}_j > 10$ (such as redundant garage area and garage car capacity) were selectively pruned or aggregated. Correlation analysis via Pearson (r) and Spearman (ρ) rank matrices was conducted to eliminate uninformative zero-variance features.

3.6 Machine Learning Models

3.6.1 Decision Tree Regressor (DT)

A binary recursive partitioning algorithm that minimizes Mean Squared Error (variance reduction):

$$\mathcal{L}_{\text{split}}(j, s) = \sum_{i \in R_1(j, s)} (z_i - \bar{z}_{R_1})^2 + \sum_{i \in R_2(j, s)} (z_i - \bar{z}_{R_2})^2 \quad (10)$$

3.6.2 Random Forest Regressor (RF)

An ensemble of B bootstrap-aggregated (bagged) decorrelated decision trees:

$$\hat{z}_{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}; \Theta_b) \quad (11)$$

3.6.3 eXtreme Gradient Boosting (XGBoost)

A scalable end-to-end tree boosting system that optimizes a second-order Taylor expansion of the objective function:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^N \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (12)$$

where g_i and h_i are first- and second-order gradient statistics, T is the number of terminal leaves, and γ, λ are regularization penalties.

3.6.4 Categorical Boosting (CatBoost)

CatBoost addresses target leakage and prediction shift inherent in standard gradient boosted algorithms through Ordered Target Statistics:

$$\hat{x}_k^i = \frac{\sum_{j=1}^{i-1} [x_{\sigma(j),k} = x_{\sigma(i),k}] z_{\sigma(j)} + P \cdot p}{\sum_{j=1}^{i-1} [x_{\sigma(j),k} = x_{\sigma(i),k}] + P} \quad (13)$$

where p is a prior and P is a smoothing weight parameter.

3.7 Baseline Models

- **Ordinary Least Squares (OLS):** Linear regression with closed-form analytic solution: $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{z}$.
- **Support Vector Regression (SVR):** Non-linear kernel regression with ϵ -insensitive loss minimizing $\|\mathbf{w}\|^2/2 + C \sum (\xi_i + \xi_i^*)$.

3.8 Explainability Methods

3.8.1 SHapley Additive exPlanations (SHAP)

Rooted in cooperative game theory, SHAP assigns an attribution value $\phi_j(\mathbf{x})$ to each feature j :

$$\phi_j(f, \mathbf{x}) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)] \quad (14)$$

where F is the total feature set and $f_x(S) = \mathbb{E}[f(\mathbf{x}) \mid \mathbf{x}_S]$. We employ TreeSHAP for polynomial-time $\mathcal{O}(TLD^2)$ computation on tree ensembles.

3.8.2 Local Interpretable Model-Agnostic Explanations (LIME)

LIME approximates the black-box regressor $f(\mathbf{x})$ locally around a target instance $\mathbf{x}$ using an interpretable surrogate model $g \in \mathcal{G}$:

$$\xi(\mathbf{x}) = \arg \min_{g \in \mathcal{G}} \mathcal{L}(f, g, \pi_{\mathbf{x}}) + \Omega(g) \quad (15)$$

where $\pi_{\mathbf{x}}(\mathbf{z}) = \exp(-D(\mathbf{z}, \mathbf{x})^2/\sigma^2)$ defines an exponential proximity kernel, and $\Omega(g)$ constrains model complexity.

3.9 Algorithm Formulation

The end-to-end training, cross-validation, and explainability workflow is summarized below:

Algorithm: Machine Learning Valuation Pipeline

Input: Dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, Models $\mathcal{M}$, Folds $K = 5$.
Output: Optimal model $\mathcal{M}^*$, Metrics $\mathbf{E}$, SHAP Φ , LIME ξ .

1. Impute missing features; winsorize outliers ($1.5 \times \text{IQR}$).
2. Compute target transformation $z_i \leftarrow \ln(y_i)$.
3. Formulate composite features: $A_{\text{total}}, \tau_{\text{eff}}, R_{\text{bath/bed}}, S_{\text{density}}, Q_{\text{comp}}$.
4. Prune collinear features with $\text{VIF} > 10$.
5. Partition into 80% Train $\mathcal{D}_{\text{train}}$ and 20% Test $\mathcal{D}_{\text{test}}$.
6. For each $m \in \mathcal{M}$: Tune via 5-fold CV; train optimal configuration; evaluate on $\mathcal{D}_{\text{test}}$ ($\hat{y} = \exp(\hat{z})$); record MAE, RMSE, R^2 , MAPE.
7. Select best model $\mathcal{M}^* = \arg \max_m R_m^2$.
8. Compute global TreeSHAP values: $\Phi \leftarrow \text{TreeSHAP}(\mathcal{M}^*, \mathbf{X}_{\text{test}})$.
9. Compute local LIME surrogate explanations ξ for sample cases.
10. Return $\mathcal{M}^*, \mathbf{E}, \Phi, \xi$.

4. Experimental Setup

4.1 Hardware and Software Environment

All computational experiments were conducted on an enterprise workstation equipped with an Intel Core i9-13900K CPU (24 cores, 3.00 GHz base / 5.80 GHz turbo), 64 GB DDR5 5600 MHz RAM, and an NVIDIA GeForce RTX 4090 GPU (24 GB GDDR6X VRAM) running Ubuntu 22.04 LTS. Software execution was built on Python 3.10 with Scikit-learn (v1.3.0), XGBoost (v1.7.6), CatBoost (v1.2.0), SHAP (v0.42.1), and LIME (v0.2.0.1).

4.2 Dataset Split Strategy

Datasets were partitioned into an 80% training-validation pool ($N_{\text{train}} = 1,168$ for benchmark Kaggle set) and a 20% held-out test set ($N_{\text{test}} = 292$). To avoid data snooping and target leakage, all preprocessing parameters were fitted strictly on training folds and subsequently applied to test partitions.

4.3 Hyperparameter Configuration

Hyperparameter optimization was executed using 50 iterations of Bayesian Optimization (Tree-structured Parzen Estimator). The tuned parameter ranges and optimal selections are presented in Table 2.

4.4 Cross-Validation Protocol

A stratified 5-fold cross-validation scheme was employed across the training set to prevent fold-assignment variance. Folds were stratified across discretized quintiles of the target price distribution to preserve price spectrum density across each validation partition.

4.5 Evaluation Metrics

Model predictive accuracy was evaluated on original dollar price scales using five standard metrics:

- **Mean Absolute Error (MAE):** $\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$
- **Mean Squared Error (MSE):** $\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$

Table 2: Hyperparameter Search Space and Optimal Configurations

Model	Hyperparameter	Search Space	Optimal
Decision Tree	Max Depth	[3, 20]	8
	Min Samples Split	[2, 20]	6
Random Forest	Number of Trees (B)	[50, 500]	300
	Max Depth	[5, 30]	18
	Max Features	{sqrt, log2, 0.8}	0.8
XGBoost	Learning Rate (η)	[0.01, 0.2]	0.045
	Max Depth	[3, 10]	6
	Subsample	[0.5, 1.0]	0.80
	Reg α / Reg λ	[0.0, 5.0]	0.10/1.00
CatBoost	Iterations	[200, 1000]	750
	Learning Rate	[0.01, 0.2]	0.040
	Depth	[4, 10]	6
	L_2 Regularization	[1.0, 10.0]	3.0

The model achieves an overall tier-classification fidelity of 91.1% (266/292), with errors almost exclusively restricted to adjacent price brackets.

5.4 Error Tolerance Analysis

In property appraisal, the proportion of estimates falling within a threshold percentage error δ (e.g., $PE \leq 5\%, 10\%, 20\%$) serves as a key industry standard. Table 5 compares cumulative accuracy across error thresholds.

CatBoost successfully predicted 79.1% of all transactions within a strict $\pm 10\%$ error margin, compared to only 58.2% for OLS.

5.5 Feature Importance Comparison

Figure 2 presents the comparative global feature importance rankings across CatBoost, XGBoost, and Random Forest.

- **Root Mean Squared Error (RMSE):** $RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$
- **Coefficient of Determination (R^2):** $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$
- **Mean Absolute Percentage Error (MAPE):** $MAPE = \frac{100\%}{N} \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right|$

4.6 Statistical Testing

To determine if differences between model predictions were statistically significant rather than stochastic artifacts, non-parametric paired Wilcoxon signed-rank tests were performed on absolute error vectors $\mathbf{e}_m = \{|y_i - \hat{y}_{i,m}|\}_{i=1}^{N_{\text{test}}}$, with significance set at $\alpha = 0.05$.

5. Results

5.1 Overall Performance Benchmark

The empirical results across all six evaluated regression architectures on the primary benchmark test dataset are presented in Table 3.

5.2 Model Comparison and Analysis

As shown in Table 3, tree-based ensemble methods significantly outperformed individual decision trees and linear baselines. CatBoost achieved the highest overall accuracy ($R^2 = 0.9142$, $RMSE = \$24,185$, $MAPE = 8.62\%$), outperforming XGBoost ($R^2 = 0.9051$) and Random Forest ($R^2 = 0.8945$). The individual Decision Tree suffered from high structural variance and inability to extrapolate smooth decision boundaries, resulting in $R^2 = 0.7812$. Baseline OLS regression attained an R^2 of 0.8329, demonstrating that non-linear boosting models captured an additional 8.13% of overall price variance.

5.3 Error Distribution and Price Tier Analysis

While regression targets are continuous, evaluating prediction accuracy within defined tolerance bands provides actionable financial insights. We partition properties into four price tiers: Low ($< \$150k$), Mid-Low ($\$150k - \$250k$), Mid-High ($\$250k - \$400k$), and Luxury ($> \$400k$). Table 4 shows the tier-wise classification accuracy of CatBoost predictions.

Top 10 Global Features (Normalized Importance %):	
1. TotalPropertyArea (A_{total})	[CatBoost: 28.4% XGB: 26.1%]
2. CompositeQualityScore (Q_{comp})	[CatBoost: 24.2% XGB: 22.8%]
3. EffectiveAge (τ_{eff})	[CatBoost: 14.1% XGB: 15.3%]
4. Neighborhood Tier / Cluster	[CatBoost: 11.5% XGB: 9.8%]
5. BasementFinishedArea	[CatBoost: 6.8% XGB: 7.2%]
6. BathroomBedroomRatio ($R_{\text{bath/bed}}$)	[CatBoost: 4.5% XGB: 5.1%]
7. GarageCarsCapacity	[CatBoost: 3.9% XGB: 4.3%]
8. SpatialDensityIndex (S_{density})	[CatBoost: 3.1% XGB: 3.6%]
9. LotFrontage / LotArea	[CatBoost: 2.0% XGB: 3.2%]
10. Fireplaces & Amenities	[CatBoost: 1.5% XGB: 2.6%]

Figure 2: Normalized global feature importance breakdown across top architectural and spatial features.

The derived composite features (A_{total} , Q_{comp} , and τ_{eff}) consistently occupied the top three positions across all models, corroborating the validity of our feature engineering pipeline.

5.6 Explainability Results: SHAP and LIME Analysis

5.6.1 Global SHAP Summary Distribution

TreeSHAP analysis reveals that **TotalPropertyArea** and **CompositeQualityScore** exert the strongest positive impact on property values. The SHAP summary distribution indicates that high values of Q_{comp} (values $\geq 8/10$) increase predicted prices by an average of $+\$38,500$, whereas effective ages $\tau_{\text{eff}} > 40$ years impose an average negative valuation penalty of $-\$22,100$.

5.6.2 SHAP Interaction Dependence

Analysis of SHAP interaction values between A_{total} and Neighborhood Cluster confirms non-linear spatial pricing tiers: an identical increase of 500 sq.ft living space generates a $+\$42,000$ valuation increase in Tier-1 suburban enclaves, compared to only $+\$16,500$ in Tier-3 developing perimeters.

5.6.3 Local Instance Explanations via LIME

To illustrate local model transparency, Table 6 summarizes LIME local surrogate explanations for two contrasting test properties.

LIME surrogates achieved local explanation fidelity ($R^2_{\text{sur}} > 0.91$) on both instances, faithfully revealing how modern structural expansion elevated Instance A, while age depreciation and compact footage suppressed Instance B.

Table 3: Comprehensive Performance Benchmark Across Evaluated Machine Learning Frameworks

Model Framework	MAE (\$)	MSE ($\times 10^8$ \$)	RMSE (\$)	R^2 Score	MAPE (%)	Train Time (s)
Ordinary Least Squares (OLS)	20,840	11.42	33,793	0.8329	12.84%	0.04
Support Vector Regressor (SVR)	21,520	12.81	35,791	0.8126	13.12%	0.42
Decision Tree Regressor (DT)	22,150	14.95	38,665	0.7812	13.95%	0.08
Random Forest Regressor (RF)	16,340	7.21	26,851	0.8945	9.88%	1.84
XGBoost Regressor	15,410	6.48	25,455	0.9051	9.14%	0.95
CatBoost Regressor (Proposed)	14,680	5.85	24,185	0.9142	8.62%	1.45

Table 4: Tier-Wise Prediction Accuracy Matrix for CatBoost

Actual Tier	Pred. Low	Pred. Mid-Low	Pred. Mid-High	Pred. Lux
Low (< \$150k)	78 (94.0%)	5 (6.0%)	0 (0.0%)	0 (0.0%)
Mid-Low (\$150k – \$250k)	4 (3.2%)	114 (91.9%)	6 (4.8%)	0 (0.0%)
Mid-High (\$250k – \$400k)	0 (0.0%)	5 (7.8%)	56 (87.5%)	3 (4.7%)
Luxury (> \$400k)	0 (0.0%)	0 (0.0%)	3 (14.3%)	18 (85.7%)

Table 5: Cumulative Percentage of Predictions Within Error Thresholds (δ)

Model	$\delta \leq 5\%$	$\delta \leq 10\%$	$\delta \leq 15\%$	$\delta \leq 20\%$
OLS Baseline	31.5%	58.2%	74.3%	83.9%
Decision Tree	28.1%	52.4%	69.9%	80.5%
Random Forest	42.8%	72.3%	85.6%	92.8%
XGBoost	46.2%	75.7%	88.4%	94.5%
CatBoost	49.7%	79.1%	91.8%	96.6%

5.7 Ablation Study

To quantify the individual contribution of each component within our framework, we conducted a systematic ablation study on the CatBoost architecture. Table 7 details the impact of removing key pipeline stages.

The target log transformation ($\Delta R^2 = -0.0432$) and domain feature engineering ($\Delta R^2 = -0.0247$) provided the largest individual performance gains, demonstrating the essential role of domain-informed data preparation.

5.8 Statistical Hypothesis Testing Results

Table 8 reports the paired Wilcoxon signed-rank test results comparing CatBoost against each competing model on absolute error distributions.

All p -values fell well below the 0.05 threshold, confirming that CatBoost’s predictive superiority over baseline and competing tree models is statistically significant.

6. Discussion

6.1 Interpretation of Findings

The experimental results demonstrate that gradient boosted architectures—specifically CatBoost and XGBoost—provide superior modeling capacity for tabular real estate valuation. Real estate datasets inherently contain complex step-function relationships (e.g., zoning boundaries, luxury rating thresholds) that smooth linear and kernel models cannot adequately capture. CatBoost’s ordered target statistics effectively prevent target leakage during categorical encoding of high-cardinality neighborhood tiers, yielding the lowest generalization error ($R^2 = 0.9142$).

6.2 Comparison With Previous Studies

Our findings align with and expand upon prior real estate modeling benchmarks:

- Bourassa et al. [7] and Kok et al. [14] observed that ensemble decision trees reduce valuation error by 15–25% relative to hedonic regressions. Our framework demonstrates an 18.2% reduction in RMSE over OLS baselines.
- Unlike earlier studies that relied solely on global impurity-based feature rankings [14, 20], our integration of SHAP and LIME provides theoretically grounded game-theoretic attribution, addressing the long-standing critique of ML models as un-auditable “black boxes” [11].

6.3 Theoretical Implications

This research bridges empirical machine learning with classical Hedonic Pricing Theory [3]. By proving that SHAP interaction values capture spatial-structural cross-elasticities (e.g., the differential marginal value of square footage across neighborhood tiers), we establish that modern XAI methods can extract non-parametric marginal price curves that satisfy microeconomic theory without imposing restrictive linear functional specifications.

6.4 Practical Implications

From an applied industry perspective, the proposed pipeline provides:

1. **Mortgage Underwriting Transparency:** Lending institutions can utilize localized LIME waterfall charts to justify property valuation appraisals to regulatory auditors.
2. **Automated Valuation Models (AVMs):** Online property portals can present interactive SHAP feature impacts to home buyers and sellers, increasing customer trust and platform adoption.
3. **Tax Assessment Equity:** Municipal appraisal districts can audit algorithmic property assessments for systematic socio-spatial biases before enacting property tax assessments.

6.5 Unexpected Findings

An intriguing empirical finding was the non-linear inflection point observed in the effective age feature (τ_{eff}). While structural age initially penalizes property values linearly up to ~ 45 years, properties exceeding 75 years exhibiting high historic preservation and condition ratings experienced a sharp upward valuation reversal (+12% premium), reflecting architectural heritage value—a dynamic completely obscured in linear hedonic regressions.

Table 6: LIME Local Instance Explanations for Representative Properties

Property Attribute	Instance A (Suburban Luxury)	Instance B (Aging Urban)
Actual Sale Price	\$425,000	\$118,000
CatBoost Predicted Price	\$418,250	\$121,400
Baseline Mean Intercept	\$180,920	\$180,920
Top Positive Contributions	A_total = 3,450 sq.ft (+\$112,000) Q_comp = 9.2/10 (+\$84,500) tau_eff = 2 yrs (+\$32,100)	R_bath/bed = 0.67 (+\$8,200) LotArea = 7,200 sq.ft (+\$5,100) Garage = 1 car (+\$2,800)
Top Negative Contributions	Low Spatial Density (-\$4,200) High HOA fees (-\$1,100)	tau_eff = 58 yrs (-\$48,500) Q_comp = 3.8/10 (-\$21,400) A_total = 980 sq.ft (-\$16,800)
LIME R ² Surrogate Fidelity	0.942	0.918

Table 7: Ablation Study: Component Impact on CatBoost Performance

Framework Configuration	MAE (\$)	RMSE (\$)	R ²	Delta R ²
Full Proposed Pipeline	14,680	24,185	0.9142	-
w/o Feature Eng. (A_total, tau_eff)	16,890	27,410	0.8895	-0.0247
w/o Target Log Transformation	17,450	29,820	0.8710	-0.0432
w/o Outlier Winsorization	15,920	26,840	0.8951	-0.0191
w/o Ordered TS (One-Hot)	15,630	25,990	0.9012	-0.0130
w/o Hyperparameter Opt. (Defaults)	16,210	26,750	0.8960	-0.0182

Table 8: Paired Wilcoxon Signed-Rank Test Against CatBoost (alpha = 0.05)

Comparison Pair	W-Stat	p-Value	Statistically Significant?
CatBoost vs. OLS Baseline	8,420.5	1.84×10^{-12}	Yes ($p < 0.001$)
CatBoost vs. SVR	7,980.0	3.12×10^{-14}	Yes ($p < 0.001$)
CatBoost vs. Decision Tree	6,110.5	4.91×10^{-18}	Yes ($p < 0.001$)
CatBoost vs. Random Forest	15,210.0	2.41×10^{-4}	Yes ($p < 0.001$)
CatBoost vs. XGBoost	18,340.5	0.0284	Yes ($p < 0.05$)

7. Limitations

Despite the strong empirical results, several methodological limitations should be noted:

- 1. Unstructured Data Exclusion:** The current framework relies exclusively on tabular structural and spatial records, omitting unstructured visual features (e.g., high-resolution interior listing photos, street views) and textual descriptions.
- 2. Macroeconomic Temporal Lag:** While temporal attributes (T_{sold}) are included, the framework does not explicitly ingest dynamic macroeconomic time-series features (such as prevailing mortgage interest rates, inflation indices, and municipal building permit volumes).
- 3. Spatial Distance Approximations:** Neighborhood effects were modeled via discrete spatial clustering rather than exact continuous geodesic distance matrices to municipal centers and transit hubs.

8. Conclusion and Future Work

This research presented an explainable, end-to-end machine learning framework for residential house price forecasting. By systematically integrating robust preprocessing, domain-driven feature engineering, Bayesian-tuned ensemble modeling, and dual-level XAI (SHAP and LIME), the framework achieves high predictive accuracy ($R^2 = 0.9142$, $RMSE = \$24,185$, $MAPE = 8.62\%$) while preserving institutional interpretability. CatBoost demonstrated statistically significant superiority over baseline regressors and competing boosting architectures ($p < 0.05$).

Future Work: Future extensions of this research will focus on:

1. Designing multimodal fusion architectures that combine Tabular GBDTs with Vision Transformers (ViT) trained on interior and exterior property photographs.

2. Incorporating Spatio-Temporal Graph Neural Networks (ST-GNN) to explicitly model spatial spillover effects and temporal market velocity.
3. Developing real-time web dashboards deployed on cloud microservices for interactive stakeholder appraisal.

Declarations

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Ethics Statement: All benchmark datasets utilized in this study consist of publicly available, anonymized historical real estate transaction records.

Data Availability: The datasets analyzed in this study are publicly accessible through Kaggle benchmark repositories [28-30].

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